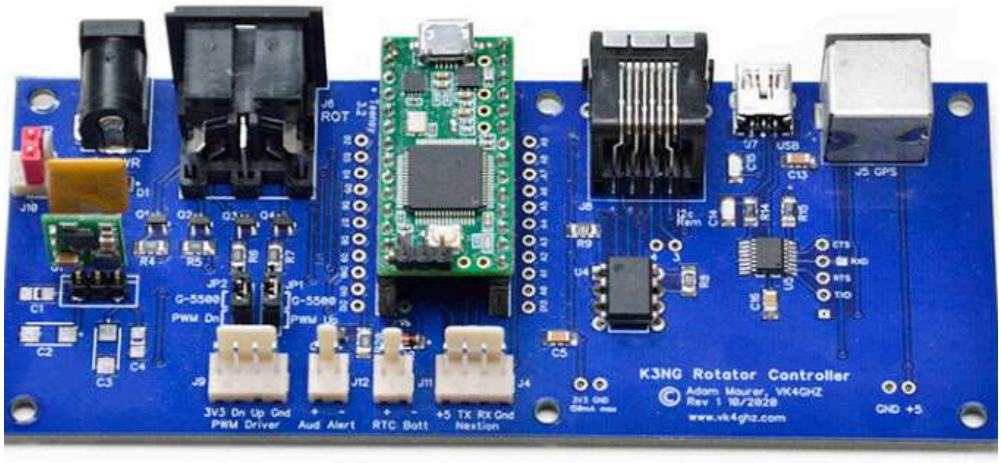
# Universal K3NG Rotator Controller System

December 17, 2020 Adam Amateur Radio, K3NG Rotator Controller, Nextion HMI, Satellites 0




**Update 11-Feb-2021:**

- **NX8048K035 5.0” Firmware Update**

A universal rotator controller project based on Anthony Good’s popular **K3NG rotator controller**.

This project features a Teensy 3.2 microcontroller and Nextion Enhanced HMI with custom user interface.

SEARCH …

VK4GHZ STORE

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$90.00

**Eltek FLATPACK2 24/48V Touchscreen Controller Kit v2 - NEW**
$150.00

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• Yaesu G-800DXA (and G-1000-DXA, G-2800-DXA) azimuth rotator control

• Yaesu G-800DXA (and G-1000-DXA, G-2800-DXA) PWM speed voltage output

• Az analogue potentiometer input

• El analogue potentiometer input

• Yaesu G-5500 compatible for CW/CCW/Up/Down control

• PWM motor driver for linear actuator elevation control

• i2c extender for remote sensors via cat5

• Spare cat5 pair pads available

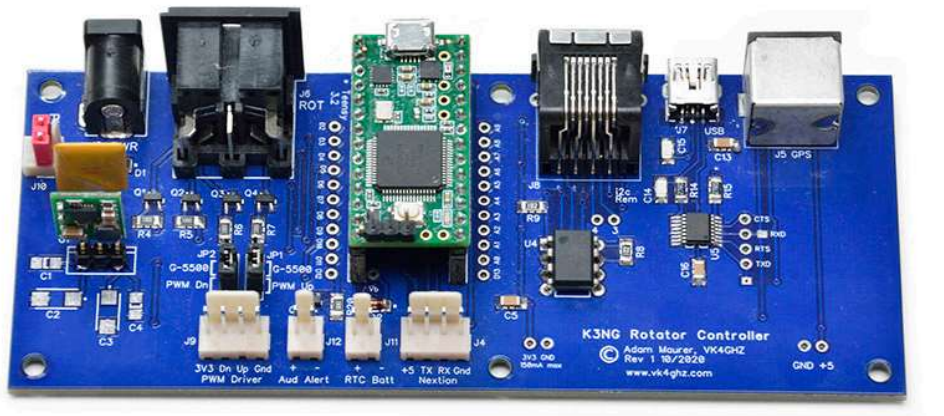
• Audible alert output

• Teensy RTC battery connection

• GPS serial data pass-through (genuine FTDI chip) to USB for time-syncing PC

• Four NO push-button switch inputs with pull-ups

**Universal Main controller board**




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**International Extra Cover**
$4.00

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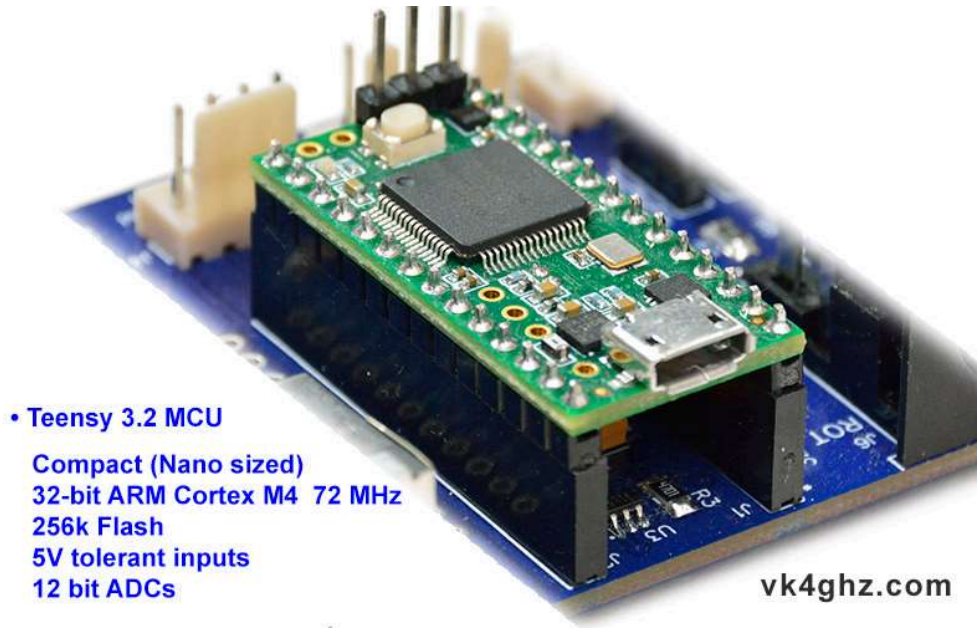
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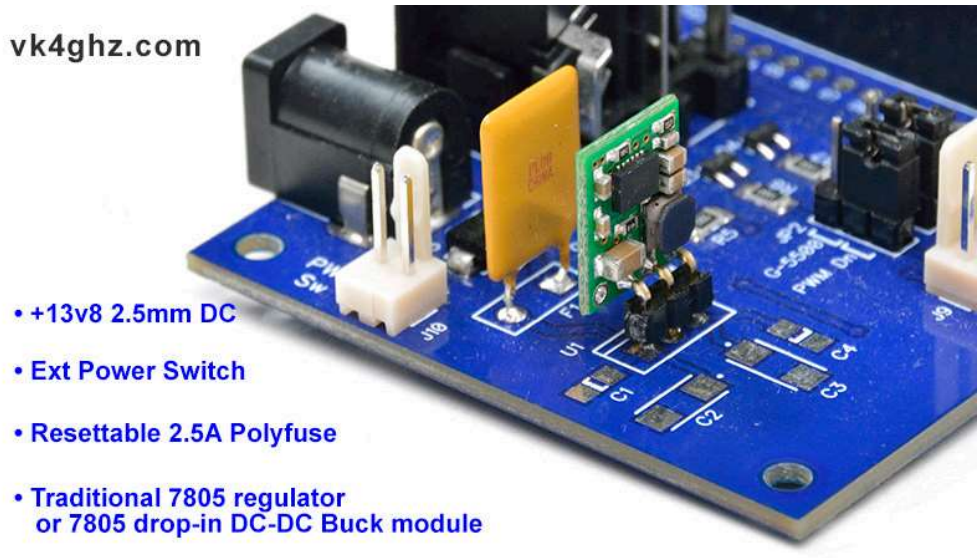
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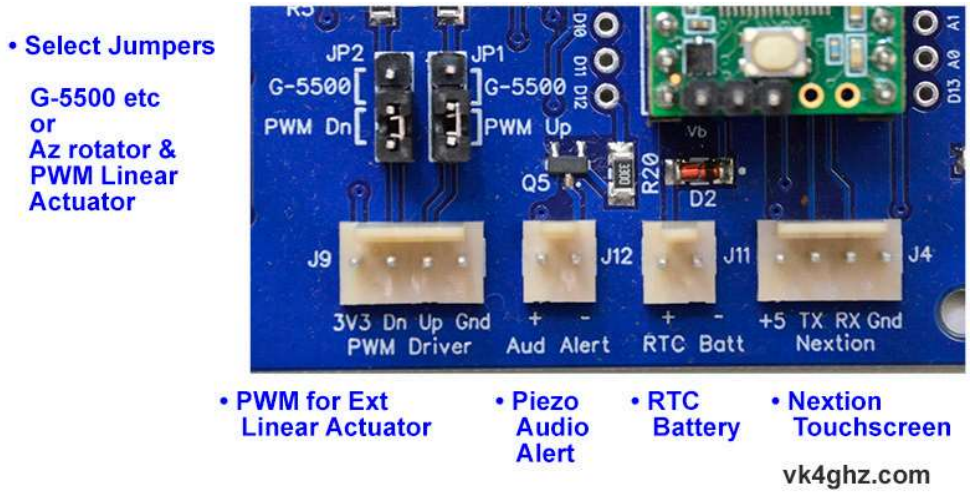
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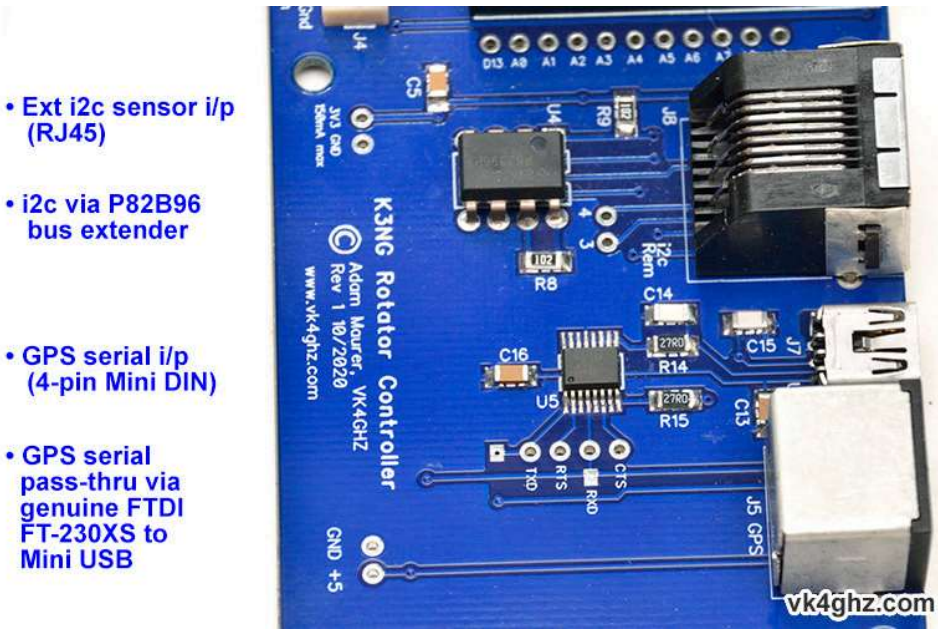
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## AMSAT LIVE STATUS

**VK Logger Forum Archive**




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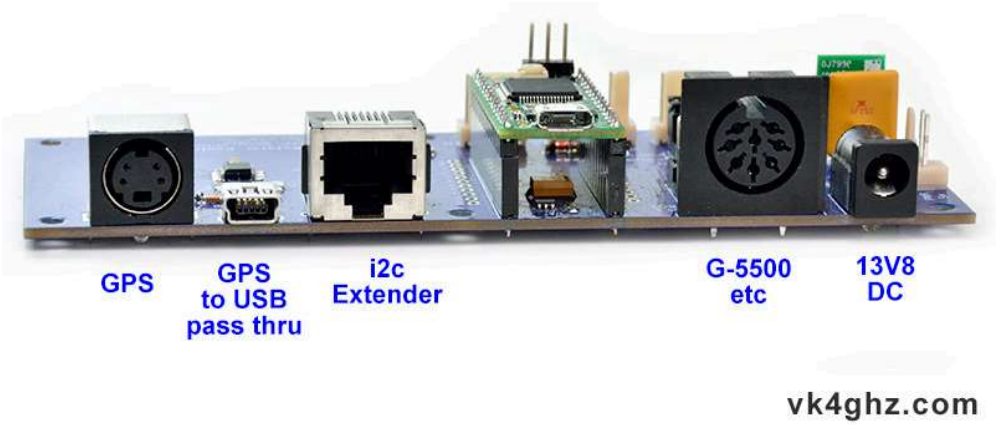


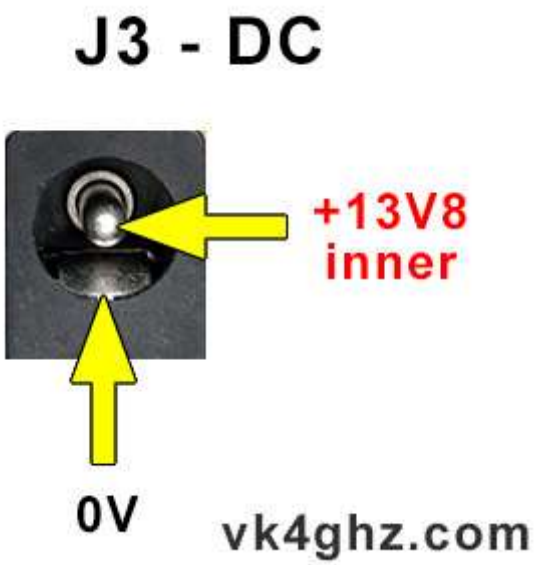
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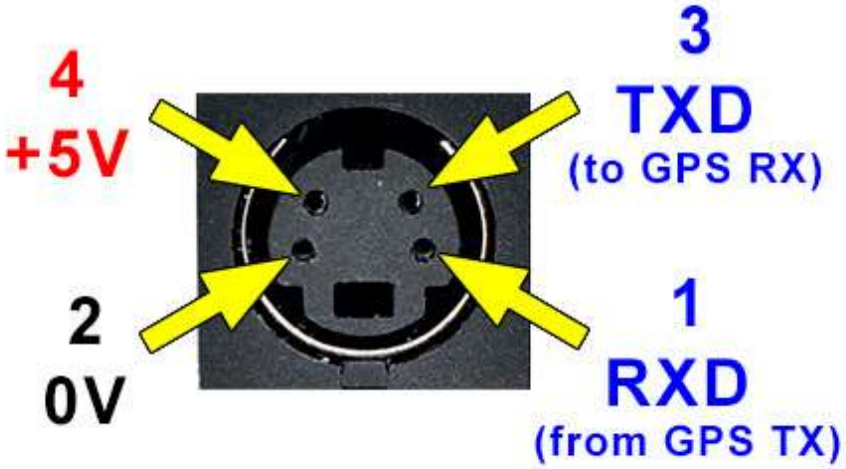
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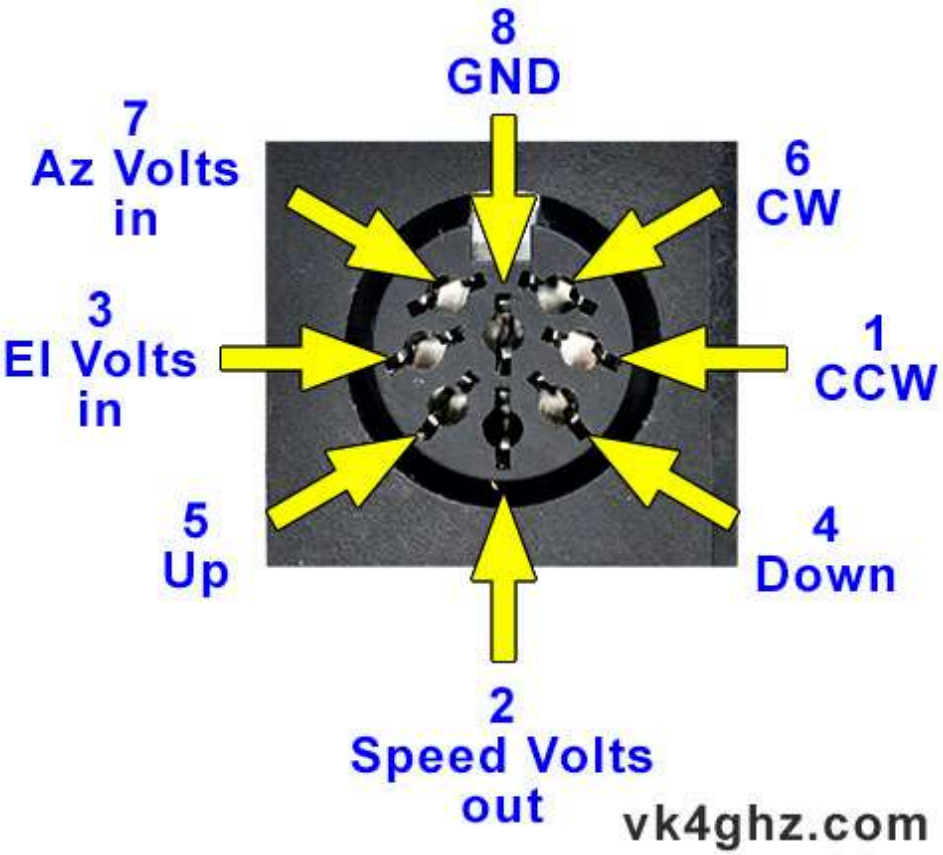
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# J5 - GPS

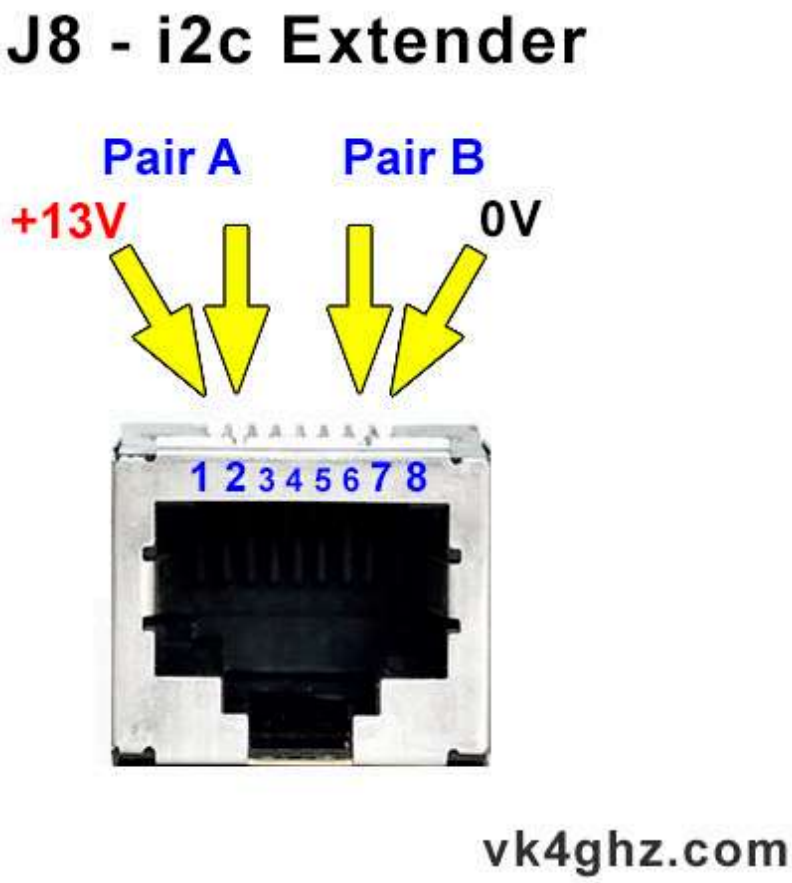



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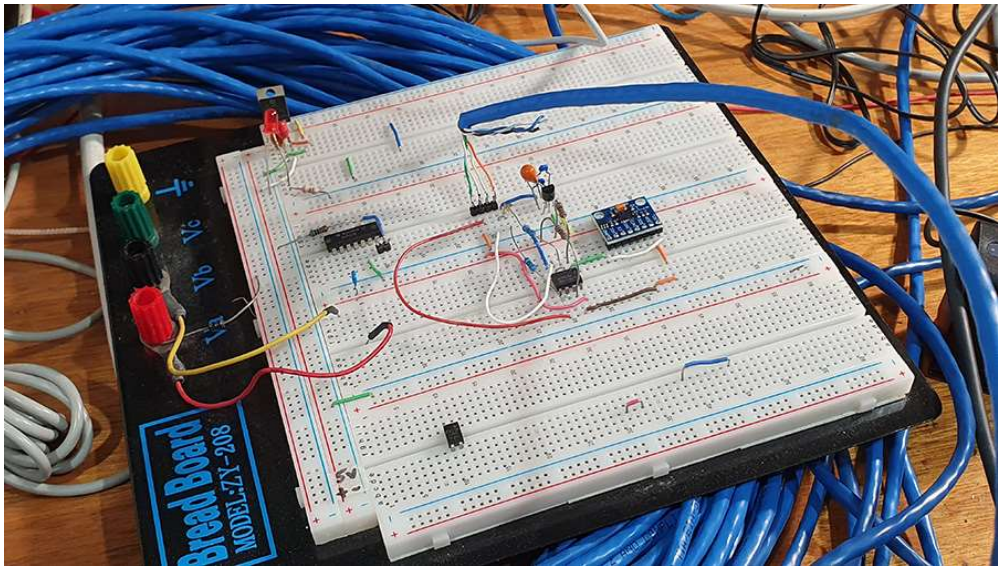
# J6 - Universal Rotator




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Development Remote Unit at the end of a 30m (100ft) run of cat5 cable.




Elevation sensor is an ADXL-345 accelerometer.  i2c extenders at both main and remote unit are P82B96 dual bi-directional bus buffers.

Primary Nextion display, one of many pages available.




Nextion screen is responsive to features set in Arduino files, so you only see what is relevant.

**Nextion HMI features:**

- Numeric Azimuth and elevation display
- Gauges Azimuth and elevation display
- CCW/CW/Up/Down rotator control
- Configurable preset positions with names
- Configurable colours/fonts
- Park button
- Go To grid square
- Co-ordinates converter – convert between lat/lon and grid square
- Configuration – Screen
- Configuration – Rotor
- Numeric keypad for entering numbers
- ABC keyboard for entering text & numbers
- Diagnostics page
- Moon & Sun tracking page
- Satellite tracking page
- Satellite selection list

- Satellite AOS page
- Satellite tracking world map

At VK4GHZ, Yaesu G-800DXA rotator for azimuth, and linear actuator (being driven by a MC33926 Motor Driver Carrier board) used for elevation control.






**Universal Schematics and PCB Overlays (pdfs)**

**Main Board** (updated 4/2/2020)

**VK4GHZ-Universal-K3NG-Rotator-Controller-Schem-Main-V12**

**PCB Top – K3NG Rotator Controller Main V3**





**PCB Bottom – K3NG Rotator Controller Main V3**

**Universal Bill of Materials**

Coming

**Remote Board** (updated 11/9/2020)



**VK4GHZ-K3NG-Rotator-Controller-Schem-Remote**

**VK4GHZ Nextion firmware for K3NG Rotator Controller available from:**




## VK4GHZ Nextion for K3NG Rotator Controller

Update 03-May-2022:  V2022-05-03 fixes typo error on About page in 7.0″ version  Nextion for K3NG Rotator Controller features: Available for: • NX8048K070 7.0″ Enhanced HMI • NX8048K050  5.0″ Enhanced HMI • NX4832K035  3.5″ Enhanced HMI • NX4024K032  3.2″ Enhanced … Continue reading

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**Project PCBs**

Will be available to purchase from the vk4ghz.com store.

More to follow, so keep checking back here.

You can find me on the Radio Artisan Groups IO:
**https://groups.io/g/radioartisan/**

Subscribe to my YouTube channel where videos will be published:

https://www.youtube.com/user/VK4GHZ

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